

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0603 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks:25

Annexure A **SCENARIO-BASED QUESTIONS**

Code of Conduct:

I X. Joshua 192312188 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

Annexure A

Short Answer Test

1. Functional Blocks of a Machine Learning System (5 Marks)

Diagram

Data Collection



Data Preprocessing



Feature Extraction



Model Training



Model Evaluation



Prediction

Functional Blocks

1. Data Collection

- Collect data from sensors, databases, websites, etc.

2. Data Preprocessing

- Remove missing values
- Remove duplicates
- Normalize data

3. Feature Extraction

- Select important attributes.
- Example: Age, Salary for loan prediction.

4. Model Training

- Train algorithm using training data.

- Algorithms: Decision Tree, Neural Network, SVM.

5. Model Evaluation

- Test model using Accuracy, Precision, Recall.

Supervised Learning

- Uses **labelled data**
- Example:
 - Email → Spam/Not Spam
 - House Price Prediction

Unsupervised Learning

- Uses **unlabelled data**
- Example:
 - Customer Segmentation
 - Market Basket Analysis

2. ML System for Disaster Management (5 Marks)

System Flow

Historical + Real-time Data



Preprocessing



Feature Selection



Neural Network Model



Prediction



Early Warning

Data Sources

- Rainfall
- River level
- Earthquake sensors
- Satellite images

Model

Use **Artificial Neural Network (ANN)** because it can learn complex patterns.

Preprocessing

- Remove missing values
- Normalize data
- Remove noise

Feature Selection

- Rainfall
- Humidity
- Seismic activity
- River water level

Training

- Split data (80% Training, 20% Testing)

Evaluation

- Accuracy
- Precision
- Recall

Applications

- Flood prediction
- Earthquake alerts
- Risk assessment
- Disaster response

3. Backpropagation in Neural Networks (5 Marks)

Steps

Input



Hidden Layer



Output



Error Calculation



Backpropagation



Update Weights

Process

Forward Pass

- Inputs pass through the network.

Error Calculation

Error = Actual – Predicted

Example

Actual = 1

Predicted = 0.8

Error = 0.2

Backward Pass

- Error propagates backward.
- Weights are updated.

Weight Update

New Weight

= Old Weight – Learning Rate × Gradient

Learning Rate

- High → Faster but unstable
- Low → Slower but accurate

Activation Functions

- Sigmoid
- ReLU
- Tanh

4. Genetic Algorithm Example (5 Marks)

Initial Population

Example	Fitness
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1	40
2	60
3	75
4	65
5	80

Selection

Choose highest fitness.

Selected:

- Example 5
- Example 3

Crossover

Parent1

Medium, High

.

Parent2

High, Medium

↓

Child

High, High

Fitness ≈ 90

Mutation

Example

High $\rightarrow$ Medium

Improves diversity.

Best Solution

Speed	Accuracy	Fitness
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High	High	90
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Mutation prevents local optimum.

5. Neural Network Classification (5 Marks)

Confusion Matrix

Dataset

TP = 2

(Two Yes predicted correctly)

TN = 1

(No predicted correctly)

FP = 1

(No predicted as Yes)

FN = 1

(Yes predicted as No)

Metrics

Accuracy

$$=(TP+TN)/(TP+TN+FP+FN)$$

$$=(2+1)/5$$

$$=60\%$$

Precision

$$=TP/(TP+FP)$$

$$=2/3$$

$$=66.7\%$$

Recall

$$=TP/(TP+FN)$$

$$=2/3$$

$$=66.7\%$$

F1 Score

$$\approx 66.7\%$$

Preprocessing

- Remove missing values

- Normalize

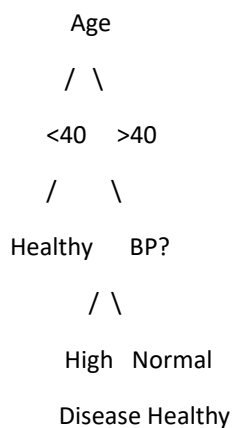
Feature Selection

- Age
- Blood Pressure
- Glucose

Improves accuracy.

6. Decision Tree Learning (5 Marks)

Diagram



Feature Selection

Entropy

Measures impurity.

Information Gain

Chooses best attribute.

Construction

- Root node
- Internal node
- Leaf node

Pruning

Removes unnecessary branches.

Preventing Overfitting

- Pruning

- Maximum tree depth
- Minimum samples

7. Smart Traffic Management ML System (5 Marks)

Inputs

- Vehicle density
- Weather
- Time
- Road type

Algorithm

Random Forest (or ANN)

Preprocessing

- Handle missing values
- Normalize

Feature Engineering

- Peak hour
- Rain indicator

Training

Train/Test split

Deployment

Real-time traffic prediction

Applications

- Signal control
- Route planning
- Congestion prediction

8. CNN Architecture (5 Marks)

Diagram

Image



Convolution Layer



ReLU



Pooling



Fully Connected



Output

Layers

Convolution

Extracts features.

Pooling

Reduces dimensions.

Fully Connected

Classification.

Feature Maps

Generated after convolution.

Backpropagation

Updates filter weights.

Example

Cat vs Dog classification.

Factors affecting performance

- Dataset size
- Learning rate
- Number of layers

9. Genetic Algorithm for Feature Selection (5 Marks)

Chromosome

Age BP Sugar BMI

1 0 1 1

1 → Selected

0 → Not selected

Fitness

Accuracy of classifier.

Selection

Best feature subsets selected.

Crossover

Combines useful features.

Mutation

Randomly changes bits.

Benefits

- Higher accuracy
- Reduced complexity
- Faster training

10. Evaluation Metrics (5 Marks)

Confusion Matrix

	Pred +	Pred -
Actual +	40	10
Actual -	5	45

TP=40

FP=5

FN=10

TN=45

Metrics

Accuracy

$$=(40+45)/100=85\%$$

Precision

$$=40/45=88.9\%$$

Recall

$$=40/50=80\%$$

F1

$$\approx 84\%$$

ROC Curve

Plots TPR vs FPR.

Applications

- Disease diagnosis
- Fraud detection

11. ANN Architecture (5 Marks)

Diagram

Input Layer



Hidden Layer



Output Layer

Layers

Input

Receives features.

Hidden

Processes information.

Output

Prediction.

Computation

Output

$$= \text{Activation}(\Sigma Wx + b)$$

W = Weight

b = Bias

Forward Propagation

Input

↓

Weighted Sum

↓

Activation

↓

Output

Prevent Overfitting

- Dropout
- Regularization
- Early stopping

12. Weather Forecasting ML System (5 Marks)

Inputs

- Temperature
- Humidity
- Wind speed
- Pressure

Algorithm

Random Forest or ANN

Preprocessing

- Clean data
- Normalize

Feature Engineering

- Humidity index
- Pressure trend

Training

Train/Test split

Deployment

Real-time rainfall prediction.

Applications

- Agriculture
- Disaster warning

13. Recurrent Neural Network (RNN) (5 Marks)

Diagram

$x_1 \rightarrow h_1 \rightarrow y_1$

↓

$x_2 \rightarrow h_2 \rightarrow y_2$

↓

$x_3 \rightarrow h_3 \rightarrow y_3$

Hidden State

Stores previous information.

Applications

- Text prediction
- Speech recognition
- Weather forecasting

Challenges

- Vanishing gradients
- Exploding gradients

Solutions

- LSTM
- GRU

- Gradient clipping

14. Genetic Algorithm for Scheduling (5 Marks)

Chromosome

Task1 Task2 Task3 Task4

Fitness

Minimum execution time.

Selection

Choose best schedules.

Crossover

Exchange task order.

Mutation

Random task swap.

Benefits

- Near-optimal scheduling
- Reduced completion time
- Efficient search

15. Performance Evaluation Metrics (5 Marks)

Sample Confusion Matrix

	Pred +	Pred -
Actual +	45	5
Actual -	10	40

TP=45

TN=40

FP=10

FN=5

Metrics

- **Accuracy** = $(TP + TN) / \text{Total} = (45 + 40) / 100 = 85\%$
- **Precision** = $TP / (TP + FP) = 45 / 55 = 81.8\%$

- **Recall** = $TP / (TP + FN) = 45 / 50 = 90\%$
- **F1-score** = $2 \times (Precision \times Recall) / (Precision + Recall) \approx 85.7\%$
- **ROC-AUC**: Measures the classifier's ability to distinguish between positive and negative classes; a higher AUC indicates better performance.

Applications

- **Spam detection**: Precision is important to avoid marking genuine emails as spam.
- **Disease diagnosis**: Recall is important to avoid missing actual patients.